

Review

Non-Noble Metal and Heteroatom Co-Doped Biochar for Cr(VI) Removal: Production, Mechanisms, and Performance Comparison

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Abstract

Hexavalent chromium (Cr(VI)) contamination in water bodies poses severe risks to ecosystems and human health due to its high solubility, mobility, toxicity, and carcinogenicity. Conventional treatment methods are often limited by high costs, secondary pollution, and inefficiency at scale. This review critically evaluates biochar (BC) as a sustainable solution for Cr(VI) removal from aqueous environments, with a specific focus on comparing the efficiency of pristine BC with that of modified BCs, including non-noble metal/BC, heteroatom/BC, and non-noble metal-heteroatom/BC. It summarizes the fundamental chemistry of Cr(VI) in aqueous environments, particularly pH- and redox-dependent speciation, bioavailability, and toxicity. Pristine BC removes Cr(VI) through electrostatic attraction, reduction to trivalent chromium (Cr(III)), surface complexation, and physisorption; however, it suffers from limited efficiency, poor surface functionality, and weak redox activity. Modification strategies overcome these limitations; for instance, iron-modified BC (Fe/BC) introduces magnetic properties, abundant reactive sites, and efficient Cr(VI) reduction, while nitrogen-doped BC (N/BC) enriches surface functional groups and improves electron transfer; Fe and N co-doped BC (Fe-N/BC) synergistically combines both advantages, achieving higher Cr(VI) removal capacities, far exceeding those of pristine BC. Characterization (SEM, FTIR, and XPS) and modelling (adsorption isotherms and kinetics and density functional theory calculations) techniques confirm the dominant chemisorption and reduction mechanisms of modified BCs. Despite promising laboratory-scale results, challenges remain in field-scale validation, long-term stability of immobilized Cr(III), nanoparticle leaching, and competitive adsorption, which are added in this review as future research directions via integration with machine learning and modelling approaches.

Keywords: nitrogen and iron co-doped biochar; chromium; adsorption; reduction; density functional theory calculations; machine learning



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1. Introduction

Chromium (Cr) is one of the most ubiquitous heavy metals (HMs) in aquatic, soil, and terrestrial ecosystems, primarily originating from a wide range of industrial activities [1,2]. Major anthropogenic sources of Cr include leather tanning, electroplating, textile dyeing, stainless-steel production, wood preservation, and pigment industries, which discharge large volumes of Cr-laden wastewater directly into water bodies or nearby soils, leading to severe environmental contamination [3]. Millions of people worldwide are estimated to be exposed to unsafe levels of Cr through drinking Cr-contaminated water, posing serious public health concerns. Unlike many organic pollutants that can be biodegraded, Cr is non-degradable and persists in the environment, accumulating in sediments and biota, thereby magnifying its ecological and human health impacts [4].

The environmental fate and toxicity of Cr are governed by its chemical speciation. It exists in two environmentally relevant oxidation states: trivalent chromium (Cr(III)) and hexavalent chromium (Cr(VI)) [5]. Cr(VI) is highly soluble, mobile, and readily transported in water environments, whereas Cr(III) tends to form insoluble hydroxides or complexes with natural organic matter (NOM), rendering it largely immobile and less bioavailable [1]. Cr(III) is recognized as an essential trace element for humans, playing a critical role in glucose, lipid, and protein metabolism by potentiating insulin action, and its deficiency is linked to impaired glucose tolerance and increased risk of cardiovascular diseases [6]. However, Cr(VI) is classified as a Group-1 human carcinogen, and inhalation of Cr(VI) dust or fumes is associated with lung cancer, while ingestion or dermal contact can cause gastrointestinal tumors; ulcers; and severe damage to the liver, kidney, and DNA [7]. Inter-conversion between Cr(III) and Cr(VI) is controlled by pH, redox potential (E_h), and presence of oxidizing or reducing agents (i.e., MnO_2 , H_2O_2 , $Fe(II)$, S^{2-} , and NOM) [8]. Under oxic and alkaline conditions, Cr(VI) predominates as chromate (CrO_4^{2-}) or dichromate ($Cr_2O_7^{2-}$) ions, both of which are potent oxidizing agents and known human carcinogens. In contrast, reducing and acidic conditions favor Cr(III) species, which have relatively low toxicity [9]. Understanding environmental risks of Cr species and designing effective treatment strategies help not only to remove total Cr but also specifically to reduce highly toxic Cr(VI) to benign Cr(III) [10].

Traditional techniques for removing Cr(VI) from contaminated water include chemical precipitation, ion exchange, membrane filtration, reverse osmosis, and electrocoagulation [11]. Although these methods can achieve high removal efficiencies under controlled conditions, they suffer from several inherent limitations that restrict their wider environmental applications [3]. Chemical precipitation, often using $FeSO_4$ or $NaHSO_4$, reduces Cr(VI) to Cr(III), followed by hydroxide precipitation. However, this method generates large volumes of toxic sludge that require costly disposal and pose secondary pollution risks [12]. Ion exchange and membrane processes are effective for concentrated streams but are expensive due to high energy consumption, membrane fouling, and need for resin regeneration, which produces concentrated brine waste [13]. Although promising, electrocoagulation requires continuous electricity and electrode replacement, making it economically unviable for large-scale treatment [14]. Moreover, many conventional methods are inefficient at low Cr(VI) concentrations (<10 mg/L), which are common in contaminated groundwater and industrial effluents after primary treatment. Therefore, there is an urgent demand for low-cost, sustainable, and highly efficient technologies for removing and detoxifying Cr(VI) while minimizing the generation of secondary waste.

Biochar (BC) is produced through the pyrolysis of biowaste feedstocks, such as agricultural residues, forestry waste, animal manure, and municipal solid waste, under oxygen-limited conditions [15,16]. BC production also generates bioenergy (syngas and bio-oil) as a co-product, improving overall energy efficiency and reducing carbon footprint. Owing to

its unique physicochemical properties, BC has a higher specific surface area (SSA), porous structure, and abundant surface functional groups (e.g., carboxyl: $-\text{COOH}$, hydroxyl: $-\text{OH}$, and carbonyl: $-\text{C}=\text{O}$) and adjustable surface charge [16]. These attributes enable BC to sequester Cr through multiple mechanisms, including electrostatic attraction, ion exchange, surface complexation, and physisorption [17]. Moreover, BC contains persistent free radicals (PFRs) and electron (e^-)-donating moieties that can directly reduce Cr(VI) to Cr(III), providing a built-in detoxification capability that is lacking in many conventional adsorbents [18]. BC application in water treatment avoids the use of hazardous chemicals that are often required by conventional methods. Spent BC can potentially be used as a soil amendment, further sequestering carbon and improving soil fertility, provided that immobilized metals are stable [19]. Life cycle assessments (LCAs) have shown that optimized BC production can achieve net negative CO_2 emissions, with a median removal of approximately 1.4 kg CO_2 -equivalent per kg of biomass [20]. These combined environmental and economic benefits make BC a highly attractive candidate for sustainable Cr(VI) removal.

Despite the demonstrated potential of pristine BC for Cr(VI) removal, its performance is often limited by its relatively low adsorption capacity, insufficient redox activity, and poor selectivity in complex environmental matrices. For instance, pristine BC typically exhibits Cr(VI) adsorption capacities <10 mg/g, which are inadequate for treating industrial effluents with high Cr loads [21]. To overcome these limitations, various modification strategies have been developed to enhance physicochemical properties of BC for Cr(VI) removal. Among these, modification of BC with non-noble metals possessing ferromagnetic properties, such as iron (Fe), nickel (Ni), and cobalt (Co), or doping with heteroatoms, such as nitrogen (N), sulfur (S), and phosphorus (P), has received particular attention due to their complementary effects on adsorption, reduction, and magnetic separability [22–24]. For instance, Fe-modified BC (Fe/BC) introduces redox-active Fe species (e.g., zero-valent iron, ZVI; magnetite; and hematite) that not only provide additional adsorption sites but also efficiently reduce Cr(VI) to Cr(III) via e^- transfer [23]. Moreover, incorporation of magnetic iron oxides (FeOx) enables facile separation and recovery of spent adsorbents using an external magnetic field, greatly enhancing reusability and reducing operational costs [25]. Heteroatom-doped BC, such as N/BC, enriches the carbon matrix with N-containing functional groups such as pyridinic, pyrrolic, and graphitic N, which improves surface polarity, e^- donor capacity, and coordination interactions with Cr(VI) species [26]. Furthermore, synergistic combination of non-noble metals and heteroatoms in BC has recently emerged as a particularly powerful approach, yielding adsorption capacities >340.9 mg/g and removal efficiencies $>90\%$ of Fe-N/BC under optimal conditions due to synergistic charge delocalization and stronger polarization [27].

Existing reviews have extensively addressed occurrence [1,3], toxicity [6,7], and conventional removal strategies for Cr(VI) [11]. In addition, emerging remediation approaches such as the use of BCs [18] and Fe-N/BCs [2] have also been reviewed for remediation of Cr-contaminated water and soil. However, persistent toxicity and widespread contamination of Cr(VI) create an urgent need for efficient and robust treatment of Cr-contaminated water using next-generation materials such as modified BCs. Given the rapidly growing literature on non-noble metal/BC, heteroatom/BC, and non-noble metal-heteroatom/BC for Cr(VI) removal, there is a need for a systematic and critical review that not only summarizes the state-of-the-art literature but also identifies mechanism insights, performance comparisons, and knowledge gaps. Therefore, this review aims to fill these critical gaps by providing systematic evaluations of pristine and modified BCs for Cr(VI) removal from water. The specific objectives are as follows: (i) to summarize the fundamental chemistry of Cr in aqueous environments, including speciation, bioavailability, and toxicity; (ii) to critically compare production, properties, and Cr(VI) removal mechanisms of pristine and

modified BCs; (iii) to quantitatively compare their adsorption capacities, influencing factors, and regeneration potential; (iv) to elucidate the underlying removal mechanisms using both experimental characterizations and computational approaches; (v) to evaluate the sustainability, economic feasibility, and environmental risks associated with modified BCs; and (vi) to identify current challenges, knowledge gaps, and future research directions, including the role of machine learning (ML) in optimizing BC design for Cr(VI) removal.

2. Cr Chemistry, Speciation, Transformation and Bioavailability

Cr(VI) is typically found in oxygenated, alkaline, and neutral pH waters as CrO_4^{2-} or $\text{Cr}_2\text{O}_7^{2-}$ ions, which increases its environmental mobility and toxicity. Cr(III) exists under reducing and more acidic conditions as hydrated cations, insoluble chromium hydroxides ($\text{Cr}(\text{OH})_x$), or complexed with NOM, whereas at neutral to alkaline pH, it precipitates as $\text{Cr}(\text{OH})_3$, decreasing its solubility and bioavailability [10]. Under dynamic *Eh* conditions, Cr(III) can be transformed to Cr(VI) via oxidation by metal oxides (i.e., MnO_x) [28]. In contrast, Cr(VI) can be reduced to Cr(III) by dissolved Fe(II), sulfide species, and microbial species [29]. pH-dependent speciation reflects the tendency of Cr to form complexes or precipitates. Cr transformation is also governed by *Eh*, microbial activity, and interactions with Fe/Mn minerals [29]. Seasonal shifts in *Eh* affect e^- donor and acceptor availability, influencing rates of Fe and sulfate reduction, which are closely related to Cr redox behavior [30]. Furthermore, MnO_x oxidizes Cr(III) to Cr(VI), whereas dissolved Fe(II)-, FeS-, and Fe(II)-bearing minerals can reduce Cr(VI) to Cr(III), creating a dynamic redox cycle. The e^- acceptors such as O_2 , NO_3^- , and NO_2^- appear to play negligible roles in Cr(III) oxidation [1]. Sediment mineralogy and ultramafic rock-derived sediments also influence Cr speciation through Fe and MnO_x [10,31]. Furthermore, NOM degradation affects redox-sensitive Fe cycling and thus indirectly influences Cr speciation and seasonal oxic–anoxic transitions in sediment, which can lead to proton exchange, acidity shifts, and variations in availability of e^- donors/acceptors that drive redox cycling of Cr [30].

Speciation-dependent bioavailability of Cr has practical implications for bioremediation and agriculture. Microorganisms capable of reducing Cr(VI) to Cr(III) can significantly decrease Cr toxicity in soils by lowering Cr(VI) concentrations and transforming them into less bioavailable forms. This biotransformation can markedly reduce bioavailable Cr(VI) fraction in soils, supporting a decrease in metal uptake by plants, thus mitigating phytotoxicity [32]. Such microbial reduction can reduce Cr(VI) bioavailability and toxicity by enhancing immobilization and precipitating less toxic Cr(III) species. Studies have shown that amendments promoting microbial Cr(VI) reduction can reduce soil Cr(VI) concentrations and bioavailable Cr by 15–50%, with corresponding decreases in plant Cr uptake and toxicity [33], indicating a strong influence of Cr speciation on bioavailability and toxicity and providing quantitative evidence for effective removal. Furthermore, speciation analysis, such as hyphenated chromatographic and mass spectrometric techniques, has improved the ability to quantify Cr species separately, enabling better risk assessment by distinguishing bioavailable and toxic Cr(VI) fraction from less bioavailable Cr(III) [3].

3. BC Production and Its Properties for Cr Removal

3.1. Feedstock Types and Their Influence on BC Properties for Cr Removal

The feedstock type used in BC production significantly influences physicochemical properties of BC and its effectiveness in removing Cr(VI) from aqueous solutions [34,35]. The choice of feedstock fundamentally dictates physicochemical properties of BC, creating a distinct trade-off between pore structure (governing physisorption and accessibility) and surface functional groups (governing chemical complexation and Cr(VI) reduction) [21,36]. Feedstocks rich in lignin tend to produce BC with a higher fixed carbon content and larger

SSA, providing more active sites for Cr(VI) adsorption than cellulose-rich feedstocks [37]. An enhanced adsorption capacity due to lignin reflects a significant increase in Cr removal efficiency, attributing to intrinsic chemical composition of feedstock [38]. Modification methods applied to BC derived from different feedstocks further improve Cr removal efficiency. For example, UV irradiation of BC substantially increases SSA and introduces abundant O-containing functional groups (–COOH and –OH), collectively enhancing surface complexation with Cr(VI). UV modification of BC increased Cr(VI) adsorption capacity by 18 times of pristine BC [21]. When comparing feedstocks post-UV modification, corn stalk-derived BC exhibited the highest sorption capacity, followed by sawdust and wheat straw BC, indicating variable feedstock influence on modification efficacy. Furthermore, chemical modification of BC feedstock with acid/alkali compounds and impregnation with metal/metal oxide nanoparticles (NPs) enhance Cr(VI) removal, owing to improved surface chemistry and pore structure [39]. Table 1 comprehensively describes the feedstock, operation conditions, and Cr(VI) removal performance of pristine BC in aqueous environments. Corn stalk and sawdust yield highly mesoporous BCs with abundant O-containing functionalities, contributing to enhanced Cr(VI) removal [40,41], whereas, coconut coir produces hard and microporous BCs with limited surface functionalities [42]. Moreover, mineral-rich sewage sludge produces macroporous BCs with diverse functional groups, but ash blocks pores, thereby affecting Cr(VI) adsorption and reduction efficiency [43]. Overall, the utilization of agricultural residues [44–46] and sewage sludge [43,47] for BC production for Cr(VI) removal addresses waste management and resource utilization.

Table 1. Feedstock, operation conditions, and Cr(VI) removal performance of pristine BC.

BC Feedstock	Operation Conditions					Removal Models		Removal Mechanisms	Cr Removal		Ref.
	Cr Concentration (mg/L)	Dose (g/L)	pH	Temperature (°C)	Time (h)	Isotherms	Kinetics		Cr(VI)	Cr _{Total}	
Bagasse pith	50	-	3	25	12	Langmuir	PSO	Physisorption, reduction, and complexation	~100%	-	[44]
Bean dregs AC	8 mmol/L	1	2	25	-	Langmuir	PSO	Reduction and adsorption	-	3.30 mmol/g	[48]
Sewage sludge	2	5	6.8	25	6	Langmuir	PSO	Adsorption, electrostatic attraction, and reduction	44%	-	[47]
Coconut coir	50	2	2	-	24	Langmuir	PSO	Adsorption-associated reduction and complexation	91.5%	-	[42]
Seaweed	1	0.5	3	30	-	Langmuir	PSO	Adsorption	64.8%	-	[49]
Chlorella	100	5	2	25	24	Redlich–Peterson	PSO	Adsorption	90.1%	-	[34]
Waste rice husks	20 µM	1	3	–20	6	-	-	Reduction and adsorption	17.9 µM	-	[50]
Biomass and plastic waste	60	0.5	5	50	40 min	Langmuir	PSO	Endothermic character and chemisorption	87.7%	-	[35]
Sewage sludge	-	4	5	-	-	-	-	Chemisorption and physisorption	<40%	92.9% Cr(III)	[43]
Oak	48	10	2	4	100	-	PSO	Reduction, adsorption, and diffusion	99%	-	[51]
Apple wood	50	10	1		72	-	PSO	Electrostatic attraction, reduction, Cr(III) complexation, and ion exchange	99.9%	-	[52]

Table 1. Cont.

BC Feedstock	Operation Conditions					Removal Models		Removal Mechanisms	Cr Removal		Ref.
	Cr Concentration (mg/L)	Dose (g/L)	pH	Temperature (°C)	Time (h)	Isotherms	Kinetics		Cr(VI)	Cr _{Total}	
Glue residue	500	2	2	25	24	Freundlich	PSO	Physisorption, electrostatic attraction, reduction, surface complexation, ion exchange, and precipitation	85%	-	[53]
Pine	50	10	2		168	-	PSO	Chemisorption	83.6%	-	[45]
Camphor tree	50	4	2	65	20	Langmuir, Temkin	Elovich	Physisorption, chemisorption, and reduction	100%	-	[46]

Note: PFO: pseudo-1st-order kinetic model; PSO: pseudo-2nd-order kinetic model.

3.2. Pyrolysis Conditions and Their Effects on Cr Removal

An increasing pyrolysis temperature typically enhances the pH, carbon and ash contents, electrical conductivity (EC), aromaticity, and SSA, all of which are important factors for HMs adsorption. Eucalyptus wood-derived BC revealed that increasing pyrolysis temperature from 450 to 950 °C increased its EC and carbon content, while decreasing the yield, volatile matter, and surface polarity. Enhanced aromaticity and surface properties at higher temperatures favor enhanced HMs removal [54]. Similarly, in spent tea leaf BC, pyrolysis at 600 °C for 2 h resulted in the highest Cd adsorption capacity (97.4 mg/g), indicating that BC produced at higher temperatures possesses greater SSA and porosity, leading to superior removal potential [55]. Residence time significantly influences BC yield and surface chemistry at lower pyrolysis temperatures. At 300 °C, increasing residence time led to a gradual decrease in yield but increased pH and adsorption capacity features; however, an extended residence time at 600 °C had negligible effects on these parameters [36]. In date seed BC prepared via slow pyrolysis at 350–550 °C for 1–3 h, longer heating times and higher temperatures elevated carbon and ash contents and enhanced surface functional groups (i.e., –COOH and phenolic moieties). Quantitative improvements in adsorption capacity with pyrolysis adjustments can be significant. UV-modified BC from various feedstocks achieved Cr(VI) adsorption capacities between 14.4 and 20 mg/g, representing a 5-fold increase compared to unmodified BC with capacities of 3.6–8.4 mg/g [21]. Although this relates to modification rather than pyrolysis, it underscores the importance of surface chemistry enhanced by pyrolysis and subsequent treatments.

3.3. Key BC Properties Relevant to Cr Removal

Cr removal is influenced by the SSA, porosity, surface functionalities, redox capacity, and structural characteristics of BC. SSA and porous structure of BC determine accessible sites for Cr(VI) adsorption. For instance, a honeycomb-structured BC derived from waste pomelo peel exhibited a high SSA of 997.5 m²/g compared to 8.8 m²/g for pristine BC, increasing Cr(VI) adsorption capacity from 194.2 to 307.3 mg/g [56]. This porous framework not only enhanced physisorption but also facilitated the photocatalytic reduction of Cr(VI) aided by PFRs present in BC. Surface chemistry, characterized by an abundance of –COOH, –OH, and C=O moieties, significantly governs electrostatic attraction, complexation, and redox interactions with Cr(VI). UV modification [21] and steam activation [48] of BCs add abundant functional groups and increase SSA, leading to an increased Cr(VI) adsorption capacity. Bean dreg ACs with different SSAs were produced by changing temperature and steam activation, and Cr(VI) adsorption capacity of ACs was positively correlated with SSA and quaternary N [48].

Eucalyptus bark BC demonstrated a sorption-reduction mechanism, where phenolic groups reduced approximately 82% of adsorbed Cr(VI) to Cr(III), with only 18% remaining as Cr(VI) on BC surface [18]. The redox capacity of BC is fundamental for transforming toxic Cr(VI) to less toxic Cr(III) and lowering Cr toxicity in treated environments. Chemical modifications, such as KOH activation, further improve pore characteristics and redox activity. KOH-modified rubberwood BC exhibited a 6-fold increase in Cr(VI) removal compared to pristine BC, attributed to its enhanced pore structure and e^- -donating capacity [57]. Similarly, KOH-activated woody BC showed SSA of 970.5 m²/g and maximum Cr(VI) adsorption capacity of 294.9 mg/g [41]. Furthermore, BC modified by incorporating nanoscale ZVI (nZVI) achieved Cr(VI) removal rates 7.6 times higher than pristine BC, with a maximum adsorption capacity of 105.65 mg/g due to synergistic physisorption (by SSA and porosity), chemisorption (by functional groups), and redox transformation (by reductive capacity of BC) [58].

4. Limitations of Pristine BC for Cr Removal

Although pristine BC is widely used for Cr removal, it exhibits several inherent limitations for effective Cr(VI) removal, particularly when compared with modified BC. These limitations are primarily related to its relatively low SSA, few reactive functional groups, limited redox activity, and inadequate porosity, which collectively constrain its adsorption capacity and removal efficiency (Table 2). For example, reed residue-derived BC showed a maximum adsorption capacity of 8 mg/g for Cr(VI), and when BC was magnetically modified by nano-Fe₃O₄, adsorption capacity increased to 9.9 mg/g [59]. Similarly, pristine BC showed a Cr(VI) adsorption efficiency of 0.4–7.6%, which was significantly lower than that of Fe/BC (50.7–98.6%) [25]. This contrast highlights the limitations of pristine BC in achieving effective removal, particularly in highly Cr-contaminated waters. Pristine BC's low abundance of reactive surface functionalities (such as -COOH, -OH) and other O-containing moieties is a critical factor limiting its Cr(VI) removal. These groups facilitate electrostatic attraction, complexation, and redox reactions, which are crucial for converting toxic Cr(VI) into less toxic Cr(III). In one study, KOH-modified BC from tropical biomass enhanced Cr(VI) removal by 6 times compared to pristine BC, which was largely attributed to increased pore characteristics and redox capacity [57]. Likewise, UV-modified BC showed an increase in O-containing groups and SSA, increasing removal capacity from 3.6–8.4 mg/g for pristine BC to 14.4–20 mg/g for UV-modified BC [21]. These modifications highlight how the lack of surface functionalities on pristine BC limits its application in Cr(VI) adsorption.

The limited redox activity of pristine BC also reduces its ability to reduce Cr(VI) to Cr(III). For instance, nZVI modification of BC enhanced Cr(VI) removal rates by 7-folds compared to pristine BC, where nZVI facilitated effective reduction alongside adsorption [58]. This underlines the limited e^- -donating functional groups and redox ability of pristine BC, which constrain its capability for detoxifying Cr via reduction. Furthermore, physicochemical structure of pristine BC, with its lower porosity and SSA, restricts available adsorption sites. A ball-milled wheat straw BC demonstrated a 3.5–9.1-fold increase in Cr(VI) adsorption capacity (52.2 mg/g) due to reduced particle size, increased SSA, and enriched O-containing functionalities [60].

Table 2. Limitations of pristine BC for Cr(VI) removal from Cr-contaminated water [61–63].

Limitations	Specific Issue(s)	Environmental Implications
Limited functional groups	Lack sufficient O-containing functional groups	Low adsorption capacity and early saturation
Low surface area and porosity	Poorly developed micropores/pores blocked by residual tars	Restricted physical trapping due to fewer available pore spaces
Selectivity deficit	Lack specialized binding sites, leading to binding of non-target and naturally occurring ions	Inhibition due to outcompeting and displacement of HMs, reducing removal efficiency
pH dependency and vulnerability	Highly sensitive surface charge of BC for pH, making it highly protonated at lower pH	Electrostatic repulsion of positively charged HMs in acidic streams
Hydrophobicity	Produced at low temperatures and often retain hydrophobic organic compounds on their surfaces	Less dispersion in aqueous solutions, limiting removal efficacy
Stability and leaching risks	Leach naturally occurring ash content, dissolved organic carbon, and HMs present in feedstocks	Introduction of secondary pollution into treated water matrix
Separation difficulties	Difficult to separate from treated water using standard sedimentation or filtration	Higher operational costs and labor intensive

5. Modification of BC with Non-Noble Metals and Heteroatoms for Enhanced Cr Removal

5.1. Non-Noble Metal-Modified BC

BC composites comprising widely accessible non-noble metals (Fe, Ni, Co, Mn, and Cu) are appealing alternatives to composites containing expensive noble metals [64]. Table 3 comprehensively describes the operation conditions and Cr(VI) removal performance of non-noble metal-modified BC in aqueous environments [65,66]. Fe is the most widely used non-noble metal for BC modification in environmental remediation. For example, modification of BC with Fe₂O₃ [67], FeCl₃ [34], nFe(0) [68], green nFe(0) [69], and magnetism introduction into BC [49] provided satisfactory results for Cr(VI) removal (Table 3). Non-noble metal modification of BC is accomplished through pre-pyrolysis impregnation; for instance, soaking biomass in Fe salts, such as FeCl₃ and FeSO₄, before pyrolysis yields a uniform Fe distribution [70]. Post-pyrolysis functionalization involves mixing pristine BC with metal solutions, followed by thermal treatment or reduction (e.g., under N₂ or H₂) [71], and addition of Fe(II)/Fe(III) to BC under alkaline conditions to form Fe₃O₄ NPs [72]. Zn-doped NiFe₂O₄–pinecone BC was synthesized via co-precipitation, and properties of Zn-doped NiFe₂O₄ were effectively modified by blending with BC (1, 5, 10, and 15%, *w/w*) that was successfully embedded with Zn-Ni ferrite NPs for Cr(VI) removal from wastewater [22]. Fe-Mn/BC was synthesized using impregnation pyrolysis, hydrothermal synthesis, co-precipitation, sol–gel synthesis, and mechanical ball milling [73]. Murtaza et al. [74] reported a 92% Cr(VI) removal efficiency using a Bi⁰-Cu⁰/BC nanocomposite synthesized by blending CuSO₄·5H₂O and 0.3 M solution of NaBH₄ and subsequent pyrolysis. Bi⁰-Cu⁰/BC exhibited superior removal efficiency compared to Cu⁰/BC (85%), Bi⁰/BC (76%), and BC (67%). Similarly, 228.8 mg/g of Cr(VI) was removed within 50 min using nano-Cu⁰/BC [75]. BC composites loaded with bio-leached nZVI via co-pyrolysis also showed 90.1% removal of 20 mg/L of Cr(VI), with 72% increase in removal capacity compared to that of pristine BC [76]. Compared with pristine BC, Fe/BC improved Cr(VI) removal

efficiency by 8.2-folds (88.8%) [77]. Chitosan (CS)-stabilized nZVI-BC composites further enhanced removal, achieving sorption capacities of 82.2 mg/g and removing >97% Cr(VI) at 2 pH [78].

Table 3. Operation conditions and Cr(VI) removal performance of non-noble metal-modified BC.

Metal-Modified BC	Operation Conditions					Removal Models		Removal Mechanisms	Cr Removal		Ref.
	Cr Concentration (mg/L)	Dose (g/L)	pH	Temperature (°C)	Time (h)	Isotherms	Kinetics		Cr(VI)	Cr _{Total}	
Bi ⁰ -Cu ⁰ /BC	25	0.5–2	4	25	2.66	Freundlich	PSO	Adsorption	92%	-	[74]
Cu ⁰ /BC	300	0.5	2	25	50 min	Sips and Langmuir	PSO	Complexation, electrostatic attraction, reduction, and pore filling	228.8 mg/g	-	[75]
Fe-modified sewage sludge BC	50	4	1	-	11.6	Langmuir	PSO, IPD	Adsorption	98.1%	-	[65]
MBC	1	0.5	3	30		Langmuir	PSO	Adsorption	82.5%	-	[49]
Fe(0)-BC and Fe ₃ O ₄ -BC	88.6	5	5	-	24	-	PFO	Adsorption	58–7%	-	[79]
nZVI/BC	5.9	0.20	3.1	25	0.5	Langmuir	PSO	Adsorption, electrostatic attraction, reduction, co-precipitation, and pore filling	95.4%	-	[80]
Fe-S/BC	5~150	10	2	45	6	Langmuir	PSO	Physisorption, electrostatic attraction, and precipitation	58.2%	-	[66]
nFe(0)/BC	100	1	4	25	24	-	-	Adsorption, direct/ortho reduction, and electrostatic attraction	64.1%	28.9% Cr(III)	[68]
Green nFe(0)/BC	50	1	2	-	8	-	PSO	Chemisorption, reduction, and co-precipitation	-	99.9%	[69]
Mixture of nZVI and dry sludge	50	4	2	25	24	Cr(VI) Langmuir and Cr _{Total} Freundlich	PSO	LFD/IPD and LFD/chemical reaction	11.6 mg/g	9.8 mg/g	[81]
γ-Fe ₂ O ₃ /BC	30	1	-	-	24	Langmuir and Freundlich	-	Adsorption (Elovich)	95.2 mg/g	-	[67]
FeCl ₃ -modified BC	100	5	2	25	24	Redlich–Peterson	PSO	Adsorption	98.4%	-	[34]

Note: MBC: magnetic BC; PFO: pseudo-1st-order kinetic model; PSO: pseudo-2nd-order kinetic model; LFD: liquid-film diffusion; IPD: intraparticle diffusion.

5.2. Heteroatom-Doped BC

Heteroatom doping (e.g., N, P, or S) modifies the electroneutral carbon backbone of BC, alters local charge distribution, introduces active surface defects, and creates functional groups that act as both adsorption sites and e[−] donors, facilitating simultaneous capture and chemical reduction of Cr(VI) to Cr(III) [82,83]. The doping and modification of BC with heteroatoms have been accomplished through pre- and post-pyrolysis doping with N-, P-, and S-rich precursors, such as urea, melamine, CS, thiourea, and chlorapatite [11,84,85]. N is the most widely used heteroatom for BC doping and modification for environmental applications and pollutant removal (Table 4) [83,86,87]. Post-pyrolysis treatment involves exposing BC to ammonia or N-containing gases at high temperatures (600–900 °C) to incorporate pyrrolic, pyridinic, and graphitic N [88]. Hydrothermal carbonization using N-containing feedstocks or additives, such as amino acids and plasma, or microwave-assisted

modification through rapid and energy-efficient surface N-grafting also modifies BC for enhanced performance [89,90]. Additionally, N-functionalized graphene aerogels composed of polyethyleneimine and polypyrrole introduced N groups, achieving high Cr(VI) adsorption capacities of 458.2 mg/g at 2 pH [91]. Cai et al. [92] reported that $\text{H}_3\text{PO}_4/\text{BC}$ removed Cr(VI) up to 55.5 mg/g via adsorption, while Gao et al. [93] used sulfide-nZVI/BC for Cr(VI) removal (141.8 mg/g) from water. N-rich activated carbon (AC) derived from agroforestry biomass with abundant amine, pyrrolic, pyridinic, and protonated N groups adsorbed Cr(VI) (255 mg/g) while reducing Cr(VI) to Cr(III), underlining critical contribution of N functionalities in both adsorption and detoxification [94]. Similarly, Ma et al. [95] reported that pristine BC derived from sludge showed a maximum Cr(VI) adsorption capacity of 8.7 mg/g compared to N/BC with 25.9 mg/g due to adsorption-reduction process, where N-containing groups on BC acted as reduction sites for Cr(VI). Collectively, these studies establish that heteroatom doping enhances BC surface chemistry, increases number of active sites, and facilitates e^- transfer, which promotes efficient Cr(VI) removal, with removal efficiencies >90%, thereby representing a highly promising and sustainable approach for effective remediation of Cr-contaminated waters (Table 4).

Dual heteroatom co-doped BCs showed superior efficiency to single heteroatom-doped BCs. Single heteroatom doping [96] alters the charge distribution of neighboring carbon atoms and localized active sites of BC, while dual heteroatom co-doping (N-S and N-P) [97,98] creates synergistic effects, such as enhanced spin density and localized defect creation, which significantly lower charge transfer resistance and boost the redox capacity (e^- donor–acceptor interactions) of BC for Cr(VI) reduction. For instance, N-S/BC sheets derived from natural biomass, with a higher SSA (1525.4 m^2/g) and pore volume (1.2 cm^3/g), and appropriate N (1.8 wt%) and S (1 wt%) co-doping showed enhanced synergistic Cr(VI) removal [84]. N-S/BC was produced by chemically modifying Fe-containing municipal dewatered sludge with $(\text{NH}_4)_2\text{S}_2\text{O}_8$ to optimize morphological structures and functional groups, achieving Cr(VI) removal efficiency of 98.7% [97]. Furthermore, pyrolyzed N-P co-doped BC from $(\text{NH}_4)_3\text{PO}_4$ -pretreated coffee shells successfully showed 95% Cr removal [98].

Table 4. Operation conditions and Cr(VI) removal performance of heteroatom-doped BC.

Heteroatom-doped BC	Operation Conditions					Removal Models		Removal Mechanisms	Cr Removal		Ref.
	Cr Concentration (mg/L)	Dose (g/L)	pH	Temperature (°C)	Time (h)	Isotherms	Kinetics		Cr(VI)	Cr _{Total}	
N/BC from NH_4Cl and bamboo	200	1	2	25	7	Freundlich	PSO	Electrostatic attraction, reduction, and complexation	-	195.8 mg/g	[83]
N/BC	50	2.5	-	25	-	Langmuir	-	Surface reactions, reduction, and immobilization	70.4 mg/g	-	[87]
N/HSBC	50	3	2	-	96	Langmuir	PSO	Redox reactions and electrostatic adsorption	99.9%	-	[96]
N-doped porous carbon	50	0.13	2	25	15	Langmuir	PSO	Spontaneous and endothermic chemisorption	193.6 mg/g	-	[86]
N/BC	30	1	-	-	4	Langmuir	PSO	Electrostatic attraction and redox reactions	100%	-	[24]
N-doped BC from bamboo	-	-	-	-	-	Langmuir	-	-	4.31 mmol/g	-	[99]
N-doped GAC (400–600 °C)	20	2	2–12	22–55	-	Sips and Redlich–Peterson	PSO	Surface, pore diffusion, and electrostatic attraction	8.7–9.8 mg/g	-	[100]

Table 4. Cont.

Heteroatom-doped BC	Operation Conditions					Removal Models		Removal Mechanisms	Cr Removal		Ref.
	Cr Concentration (mg/L)	Dose (g/L)	pH	Temperature (°C)	Time (h)	Isotherms	Kinetics		Cr(VI)	Cr _{Total}	
N-doped BC from CS gel	200	1	2	25	24	Langmuir	PSO	Chemisorption, electrostatic effect, and physisorption	190.4 mg/g	-	[101]
O-N/BC	5	2	3	25	20	Langmuir	PSO	Surface and intraparticle diffusion	51.5–81.8% (28.3 mg/g)	-	[102]
S-N/BC	100	1	2	25	24	Langmuir	PSO	Electrostatic attraction, reduction, and complexation	98.7%	-	[97]
N-containing BC	150	1	2	25	24	Langmuir	PSO	Pore adsorption, electrostatic attraction, and partial chemical interaction	86.2%	-	[103]
N-doped AC	500	1	1	25	-	-	PSO	Chemisorption and physisorption	386.2 mg/g	-	[38]
N-rich AC	50	0.5	2	-	-	Langmuir	PSO	Adsorption-coupled reduction	97.1 mg/g	-	[94]
N-P/BC	300	1	3	-	4	Sips	-	Electrostatic attraction and redox reaction	-	95%	[98]
N-S/micro-mesoporous C sheet	10	40	Unadjusted	25	-	Langmuir	PSO	Physisorption, electrostatic attraction, in situ reduction, and surface chelation	-	81.5%	[84]

Note: PSO: pseudo-2nd-order kinetic model.

5.3. Non-Noble Metal and Heteroatom Co-Doped BC

To address the inconsistencies of non-noble metal-modified BCs and heteroatom-doped BCs, the production of non-noble metal and heteroatom co-doped BCs could be a practicable option that triggers a powerful structural and electronic synergy in BCs. Heteroatom and metal co-doped BCs significantly enhance capabilities for Cr(VI) removal through synergistic effects of positive surface charges and functionalities from heteroatoms and e^- -dense centers and adsorption active sites from metal NPs (Table 5) [104,105]. Co-doping boosts synergistic redox sites, e^- transfer, and Cr(VI) adsorption and reduction [106,107]. For example, N-doped wheat straw BC loaded with nZVI achieved Cr(VI) adsorption capacity of 41.3 mg/g and maximum monolayer adsorption capacity of 100.9 mg/g, wherein heteroatom doping with nZVI promoted dual pathways: partial adsorption of Cr(VI) on BC surface and predominant reduction of Cr(VI) to Cr(III) mediated by Fe(0) and Fe(II) species [26]. Similarly, nanocomposites combining magnetic Fe₃O₄ NPs with N-containing polymers, such as CS and polypyrrole, exhibited maximum Cr(VI) adsorption capacities of approximately 105 mg/g, with adsorption coupled with chemical reduction [108]. Magnetic N/BC facilitated Cr(VI) removal through N functionalities and metallic NPs, wherein metal species provide reductive sites, and BC's carbon matrix facilitates e^- transfer, with heteroatom dopants potentially enhancing these processes indirectly by modifying carbon structure [76]. Polyethylenimine (PEI) magnetic BC (MBC) also showed higher Cr(VI) removal capacity of 183 mg/g [109]. Thus, synergy between heteroatom doping and metal modification not only increases available active sites but also promotes e^- transfer, which is crucial for redox transformation of Cr(VI) to

Cr(III). This results in chemisorption-dominated removal with high capacity and rapid kinetics, making Fe-N/BC promising for Cr removal [110].

Metal and heteroatom co-doped BCs showed higher Cr(VI) removal efficiency than pristine BCs, owing to homogeneous dispersion of metals and heteroatoms on BC surface [111]. For instance, removal capacity of Fe-N/BC (105.6 mg/g) outperformed that of pristine BC (42.3 mg/g) [112], and removal efficiency of Fe-N/BC (99.9%) was higher than that of Fe/BC and N/BC because of positively charged N groups on BC surface [106]. Fe-N/BC also reported a 93.8 mg/g Cr(VI) removal [11], signifying effective synergistic roles of Fe, N, and BC in Cr(VI) removal. In addition to Fe-N co-doping into BC, FeS incorporation into BC also provided effective Cr(VI) removal. FeS-N/BC reduced Cr(VI) in solution via precipitation [113], and nFeS/N-BC with a higher adsorption capacity (373.8 mg/g) removed Cr(VI) via adsorption and reduction [114]. Ball-milled Fe-S/BC synthesized via grinding Fe(0), S(0), and BC showed enhanced Cr(VI) removal efficiencies (76.9–99.4%) and P-recovery, demonstrating enhancing and synergistic effects of Fe-S/BC for HMs removal and nutrient recovery [66].

Table 5. Operation conditions and Cr(VI) removal performance of non-noble metal and heteroatom co-doped BC.

Metal and Heteroatom Co-Doped BC	Operation Conditions					Removal Models		Removal Mechanisms	Cr Removal		Ref.
	Cr Concentration (mg/L)	Dose (g/L)	pH	Temperature (°C)	Time (h)	Isotherms	Kinetics		Cr(VI)	Cr _{Total}	
ZVI/N-BC	40	0.5	5.6	25	4	-	-	Electrostatic attraction, ion exchange, and redox reaction	97.6%	-	[115]
nFeS/N-BC	-	-	-	-	-	Langmuir	PSO	Adsorption and reduction	373.8 mg/g	-	[114]
PEI grafted N-MBC	100	0.5	2	25	-	Freundlich and Sips	-	Pore filling, electrostatic attraction, ion exchange, precipitation, and complexation	183 mg/g	-	[109]
Fe-N/BC (300–900 °C BCs)	100	1	3.73	25	24	Langmuir	PSO	Electrostatic attraction, reduction, and surface complexation	93.8 mg/g	-	[11]
Fe-N/BC	50	1	2	25	10 min	-	-	Adsorption, reduction, and Cr(III) complexation	99.9%	-	[106]
N/MBC	50	0.5	2	25	12	Langmuir	PSO	Chemisorption	133.3 mg/g	-	[104]
FeS-N/BC	100	0.5	5.4	30	0.5	Langmuir	PFO	Direct solution reduction and precipitation	100%	-	[113]
Fe/N-BC	200	0.5	2	25	2	-	-	Chemisorption	340.9 mg/g	-	[105]
Fe-N/PBC	50–200	0.5	2	25	2	Langmuir and Sips	Avrami	Reduction, electrostatic attraction, complexation, pore filling, and ion exchange	340.9 mg/g	-	[27]
Fe-N/BC	10	0.2	2	25	2.5	-	-	Electrostatic attraction and H ⁺ and e ⁻ transfer	97.7%	-	[111]
ZnCl ₂ -activated N/BC	50	1	-	-	24	-	-	Coordination interaction, electrostatic attraction, and chemical reduction	81.1%	-	[116]

Table 5. Cont.

Metal and Heteroatom Co-Doped BC	Operation Conditions					Removal Models		Removal Mechanisms	Cr Removal		Ref.
	Cr Concentration (mg/L)	Dose (g/L)	pH	Temperature (°C)	Time (h)	Isotherms	Kinetics		Cr(VI)	Cr _{Total}	
N-doped GO 3-D hydrogel on MBC	100	20	2	25	0.5	Langmuir	PSO	Electrostatic attraction, pore filling, coordination, and redox reaction	-	350.4 mg/g	[107]
N-F/BC	50	-	2	25	16	Langmuir	PSO	Chemisorbed and reduction-adsorption	-	98.6%	[112]

Note: PFO: pseudo-1st-order kinetic model; PSO: pseudo-2nd-order kinetic model.

6. Physicochemical Properties of Modified BC

6.1. Increased SSA and Pore Development

Compared to pristine BCs, metal-modified BCs generally exhibit significantly enhanced SSA and pore structure development of mesopores and micropores, along with introduction of metal oxide active sites, directly contributing to their improved adsorption performance [39,117]. Corn straw BC modification by FeCl₃ substantially improved SSA and pore size, promoting Cr(VI) adsorption [39]. Likewise, Fe/BC prepared via co-precipitation demonstrated increased pore structures accompanied by enriched C=O, -COOH, and metal hydroxyl functional groups, which collectively enhanced Cr(VI) removal capacities [118]. FeOx impregnation into brown seaweed-derived BC increased SSA and porosity, thereby enhancing adsorption sites for HMs removal compared to pristine BC [119]. UV modification increases BC's SSA and functionalities, which is partly attributed to enhanced microporous and mesoporous structures that provide more accessible adsorption sites for enhanced Cr adsorption capacity [21].

6.2. Introduction of Surface Charge and Specific Functional Groups

BC modification with metals significantly enriches BC surface with active adsorption sites. Fe in BC predominantly exists as FeOOH and Fe₂O₃, which introduces additional active sites, facilitating strong adsorption interactions [117]. Similarly, introduction of FeOx into magnetic kenaf fiber BC not only improved textural properties (i.e., SSA and pore volume) but also increased O-containing surface functional groups. This led to formation of inner-sphere complexes between HM ions and O-containing groups, markedly improving adsorption (47.9 mg/g) [120]. Similarly, nano-FeOx/BC synthesized from microalgae and Fe(II) salt showed that FeOx incorporation introduced additional O-containing groups, boosting adsorption capacity for dyes [121]. N doping also enriches the surface chemistry of BC by introducing a variety of N-containing functional groups (i.e., pyrrolic and pyridinic N). N-doped BC produced from birch tree waste using melamine showed an increased N content (0.5–5.4%), mesopores, and N functional groups, improving adsorption efficiency and surface chemical reactivity [122]. Modified BCs also exhibit enhanced electrostatic attraction towards Cr due to introduction of positively charged metal species on their surfaces. Fe/BC achieved Cr(VI) removal efficiencies ranging from 50.7 to 98.6% compared to pristine BC (0.4–7.6%) due to positively charged surfaces enhancing electrostatic attraction [25]. Non-noble metal and heteroatom doping also influences electrostatic attraction; N functionalities on BC surface exhibited >81% Cr(VI) removal efficiency due to synergistic adsorption through different mechanisms [116]. Moreover, N/BC loaded with nZVI demonstrated a maximum Cr(VI) adsorption capacity of 41.2 mg/g under acidic conditions, where N doping improved surface charge characteristics to favor electrostatic attraction alongside chemical reduction [26].

6.3. Magnetism of BC

Non-noble metal-modified BC has enhanced magnetic properties due to successful deposition of metal oxides and zero-valent metals on its surface, enabling efficient magnetic separation of BC composites from aqueous environments, thereby facilitating their recovery and reuse [123]. For instance, corn straw-based MBC showed significantly improved Cr(VI) removal efficiencies, largely attributed to FeOx or ZVI particles deposited on their surfaces that introduced magnetism, which allowed for facile separation and regeneration [25]. Perera et al. [124] reported 16% Fe/BC recovery after first desorption cycle during Cr(VI) removal. N doping coupled with Fe modification produces BCs that benefit from both enhanced magnetic separation and improved adsorption via additional active sites [111]. N-Fe/BC demonstrated improved adsorption performance and better separation through magnetism, stemming from presence of FeOx alongside N-enriched surface functionalities [125]. This dual modification thus combines the magnetic advantages of metal species with the enhanced chemical activity of heteroatom species, promoting overall BC material efficiency and reusability [118]. Furthermore, Fe-N/BC possesses enhanced magnetic properties and higher reusability (91%), owing to enhanced surface functionalities, structural stability, and adsorption capacities over multiple cycles [126].

6.4. Catalytic Properties

BC's role as an adsorbent for HMs removal has been comprehensively elaborated in the literature; however, its role as a catalyst lacks in-depth exploration. Modified BCs exhibited significantly enhanced catalytic degradation compared to pristine BCs for Cr(VI) removal [34,49]. Modification introduces FeOx or ZVI onto the BC surface, acting as active sites that facilitate redox reactions to reduce Cr(VI) to Cr(III). Modified corn straw-based Fe/BC demonstrated Cr(VI) removal efficiencies ranging from 50.7% to 98.6%, far exceeding 0.4% to 7.6% range observed for pristine BC, highlighting an increase in degradation performance. The improved catalytic effect is largely attributed to electrostatic attraction and enhanced redox activity stemming from deposited metal species [25]. Similarly, Fe-modified honeysuckle-derived BC showed spontaneous and endothermic adsorption with enhanced kinetic performance for Cr(VI) removal, further corroborating that metal modification significantly accelerates catalytic degradation through synergistic mechanisms involving physisorption and chemical reduction [127]. Moreover, rapid microwave-synthesized nano-magnetite-modified BC demonstrated a higher maximum adsorption capacity (9.9 mg/g) than pristine BC (8 mg/g), displaying faster catalytic removal rates attributed to nano-Fe₃O₄ active sites [59]. N/BC also enhances catalytic degradation capabilities of pristine BC by introducing pyrrolic and pyridinic N, which modify surface chemistry and electronic properties, thereby promoting chemical interactions with contaminants. N-doped wheat straw BC loaded with nZVI exhibited notable Cr(VI) adsorption, wherein N doping facilitated stronger coordination interactions and electrostatic attraction, intensifying catalytic reduction and adsorption [26]. The presence of metal and heteroatom species in BC increased the density of redox-active Fe species and surface chemical functionalities, notably N groups, contributing to faster e⁻ transfer and greater adsorption affinity toward Cr(VI) as well as chemical reduction of Cr(VI) to Cr(III). Fe-P/BC showed several-fold increases in Cr(VI) removal compared to pristine BC and Fe/BC, with faster adsorption kinetics, underscoring chemisorption and catalytic transformation promoted by co-modification of BC with Fe and P [85].

7. Influencing Factors for Cr(VI) Removal

Although numerous environmental factors, including pH, BC dose, Cr(VI) concentration, co-existing ions, NOM, and suspended solids, significantly influence Cr(VI) removal

by BCs. Presence of suspended solids enhance the adsorption capacity of Cr(VI) due to higher sorption sites [128]. Similarly, NOM, i.e., humic and tannic acid species (5 mg/L), strongly enhanced Cr(VI) removal by 73.2 and 84.4%, respectively, due to increased interception and gravitational sedimentation, which promoted the adsorption of Cr(VI) onto NOM [129]. Since extensive research has been done only on the effects of pH, BC dose, Cr(VI) concentration, and co-existing ions on Cr removal, only these factors are discussed in this section (Figure 1).

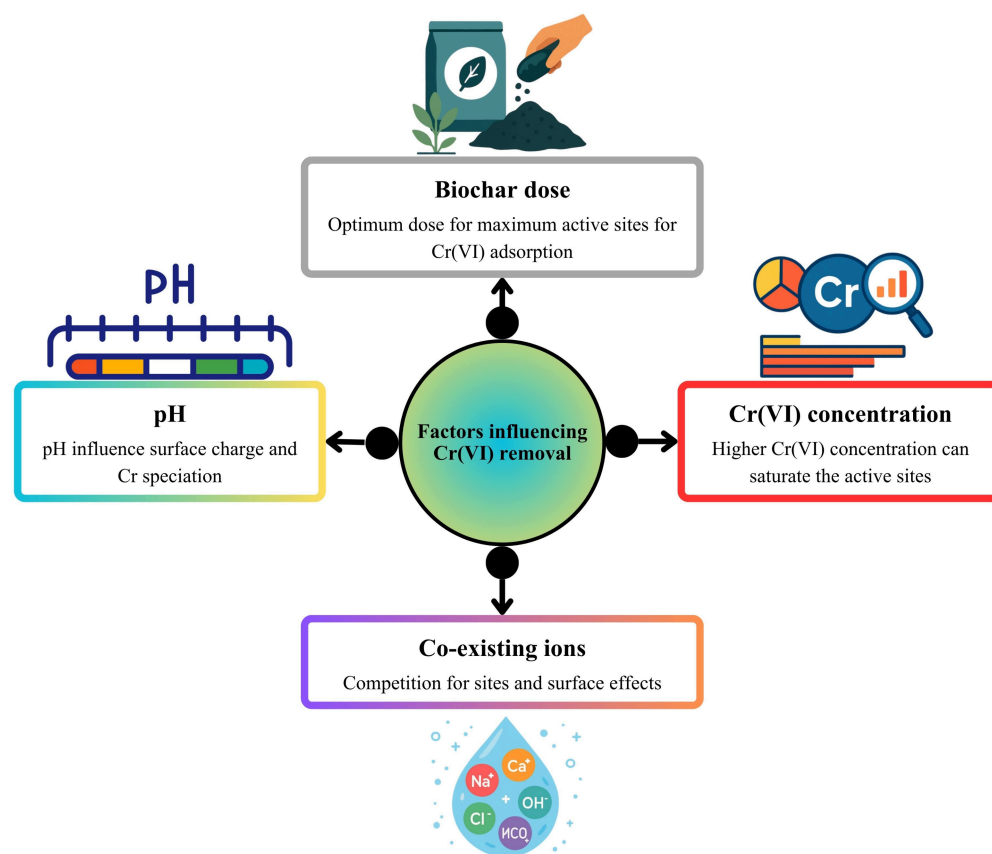


Figure 1. Factors influencing Cr(VI) removal by BCs.

7.1. pH

Solution pH strongly affects the Cr removal efficiency of BC, with acidic conditions generally enhancing removal performance due to protonation of surface sites and increased positive surface charges, which intensify electrostatic attraction towards negatively charged Cr(VI) oxyanions [112]. Fe/BC exhibited Cr(VI) removal efficiency of 98.6% under acidic conditions ($\approx 2\text{--}3$ pH) compared to 10% Cr(VI) under neutral or alkaline conditions. This is due to a decreased positive charge and protonation and increased competition from OH⁻ ions at a higher pH, reducing electrostatic attractions and adsorption capacity [25]. Similarly, optimal Cr(VI) adsorption with Fe/BC occurred at pH 2, illustrating that FeOx renders BC particularly effective in acidic media through enhanced surface charge and redox activity, achieving higher Cr removal efficiencies that sharply decrease with an increasing pH [39]. N/BC is also affected by pH conditions due to enhanced e⁻ donor capacity and N functional groups that interact strongly with HMs and influence surface charge. For instance, N/BC achieved an 81% Cr(VI) removal rate, outperforming pristine BC, under acidic conditions (≈ 2 pH), attributed to improved electrostatic attraction between protonated N-containing groups and anionic Cr species, as well as coordination interactions and chemical reduction. However, as pH increases, protonation decreases,

leading to diminished electrostatic effects and adsorption capacity. Such pH-dependent variations underscore the role of heteroatom functionalities in tuning surface charge and reactive site availability, augmenting Cr removal in acidic environments [26]. For the pronounced dependence of Cr removal with metal and heteroatom co-doped BCs on solution pH, Fe-N/BC showed an adsorption capacity of >41 mg/g at pH 2, reflecting boosted electrostatic attraction due to protonated functionalities and enhanced reduction kinetics by Fe species [130]. It is worth noting that Cr(VI) adsorption at $\approx 2\text{--}3$ pH is technically optimal as most BC surfaces acquire positive surface charges, leading to strong electrostatic attractions and higher Cr(VI) removal efficiency (Tables 1, 3, 4 and 5) [112]. But maintaining these pH conditions proves economically impractical for large-scale treatment due to high acid consumption for pH adjustment, severe equipment corrosion requiring costly materials, strict safety protocols, expensive post-treatment neutralization, and difficult adsorbent regeneration.

7.2. BC Dose

Increasing BC dose increases Cr(VI) removal efficiency by providing more SSA and active sites for adsorption and reduction. Increasing nZVI/BC dose from 0.2 to 8.0 g/L increased Cr(VI) adsorption rates from 17.7 to 87.7% due to more active sites for adsorption [68]. Similarly, N/BC showed an increasing trend of Cr(VI) removal when the BC dose increased from 0.2 to 1%, with maximum removal accomplished at 0.5 g/L [113]. Increasing BC dose also enhances removal by supplying more metal and heteroatom-functionalized sites for electrostatic attraction and chemical interaction with Cr ions [116].

7.3. Cr(VI) Concentration

Initial Cr(VI) concentrations affect Cr removal efficiency due to availability of active adsorption sites and competition effects [81]. Fe/BC demonstrated significantly enhanced Cr(VI) removal, which was largely attributed to an increased number of adsorption sites formed by FeOx on BC surfaces [131]. An increase in initial Cr concentration resulted in a decrease in removal, indicating that available adsorption sites could be saturated at higher Cr(VI) concentrations. This exemplifies that while increasing BC dose enhances efficiency, very high Cr(VI) concentrations may overwhelm adsorption sites, thus reducing removal efficiency per unit mass. ZnCl₂-activated N/BC exhibited a dependency on BC dose and Cr concentration for their removal performance, mediated by enrichment of N-containing functional groups that create additional active sites and modify surface charge. As the initial Cr(VI) concentration increased, removal efficiency decreased due to finite adsorption capacity and site saturation [116]. Additionally, non-noble metal and heteroatom co-doping promote synergistic coordination interactions and chemical reduction mechanisms, maintaining elevated Cr adsorption even at relatively high Cr(VI) concentrations, demonstrating an improved capacity over pristine BC.

7.4. Co-Existing Ions

Co-existing ions affect Cr removal efficiency of BC through competitive electrostatic attraction and competition for adsorption sites [83,111]. Main competing co-existing anions can be sulfate (SO₄^{2−}), phosphate (PO₄^{3−}), carbonate (CO₃^{2−}), bicarbonate (HCO₃[−]), chloride (Cl[−]), and nitrate (NO₃[−]) to reduce the Cr(VI) adsorption performance of modified BCs in real water environments [132]. Because PO₄^{3−} and Cr(VI) share adsorption mechanisms involving ligand exchange and electrostatic attraction at Fe sites, co-existing anions such as PO₄^{3−}, SO₄^{2−}, or CO₃^{2−} can affect Cr(VI) removal efficacy [133]. Liu et al. [134] checked the Cr(VI) removal efficiency of Fe carbide–N-doped carbon nanotubes in the presence of SO₄^{2−}, HPO₄^{2−}, Cl[−], Br[−], and SO₄^{2−} and reported that Cl[−], Br[−], and SO₄^{2−} promoted Cr(VI) removal, while HPO₄^{2−} decreased Cr(VI) removal efficiency by 65.3%.

Lignin-derived N-doped AC prepared via one-pot production showed enhanced SO_2 , NO, and Cr(VI) removal performance due to better structure development of ACs [38]. N/BC used for Cr(VI) removal reported that the presence of Cd(II) and Cu(II) did not significantly affect Cr(VI) removal, but AsO_4^{3-} and SeO_4^{2-} reduced the Cr(VI) adsorption capacity of BC [115]. N-functional groups promote coordination interactions and enhance electrostatic attraction of Cr species, improving resistance to interference by cations and anions. Synergy between metal species and heteroatom functionalities creates more diverse set of active and selective adsorption sites, enhancing chemical reduction and coordination mechanisms that are less susceptible to competitive adsorption than purely physical or electrostatic sites. Fe-N/BC demonstrated higher Cr(VI) adsorption capacities, attributing to combined adsorption and reduction facilitated by both Fe and N sites [26]. ZnCl_2 -activated N/BC was less affected by co-existence of Cr(VI) and Pb(II) because N-containing surface groups can modify adsorption selectivity and affinity [116]. Although direct quantitative data on the effects of co-existence of HMs on heteroatom-doped BCs for Cr removal are limited, studies on multi-ion systems indicate that N doping enhances affinity towards Cr(VI). Further research is needed to clarify the highly situation-specific selectivity of non-noble metal and heteroatom co-doped BCs for Cr(VI) removal in the presence of competing cations and anions.

8. Experimental and Computational Elucidation of Cr(VI) Removal Mechanisms by BCs

8.1. Removal Mechanisms of Cr(VI)

Non-noble metal and heteroatom co-doped BCs employ six major mechanisms, such as electrostatic attraction, chemical reduction, surface complexation, co-precipitation, ion exchange, and pore filling, for Cr(VI) removal (Figure 2). At acidic pH (≈ 2 –3), abundant H^+ significantly enhances reduction potential, accelerating e^- transfer from O-containing functionalities, doped heteroatoms, or zero-valent metals for Cr(VI) removal [112]. While chemical reduction is the key Cr(VI) transformation mechanisms [106], it is electrostatic attraction that acts as a crucial driving force at ≈ 2 –3 pH by generating a localized high-concentration environment on BC surfaces for subsequent reduction, surface complexation, and co-precipitation [83].

8.1.1. Electrostatic Attraction

Electrostatic attraction is a key mechanism for Cr removal from water (Figure 2). Metal-modified BCs possess metal species (such as ZVI and FeOx) introduced onto the BC surface and confer a positive surface charge, especially under acidic conditions, which strongly attracts negatively charged Cr(VI), leading to enhanced electrostatic attraction, alongside adsorption and reduction, with positively charged Fe/BC surfaces effectively capturing negatively charged Cr anions [76]. Electrostatic attraction is strongly pH-dependent and is a dominant contributor to HMs removal. Tariq et al. [135] reported the highest Cr(VI) adsorption (75 mg/g) with CuO- and MnO-impregnated BC (Mn-Cu/BC), followed by Cu/BC (59.6 mg/g), Mn/BC (44.9 mg/g), and pristine BC (37.4 mg/g), accomplished through H^+ bonding, electrostatic attraction, complexation, and redox reactions. Similarly, ZnCl_2 -activated N/BCs utilize electrostatic attraction as an essential part of their Cr removal, which is enhanced by the introduction of N functionalities that modify the surface charge and e^- -donating characteristics and increase positive charge density under acidic conditions, facilitating stronger electrostatic attraction for Cr(VI) species [116]. The synergy of Fe-N/BC further amplifies electrostatic attraction for Cr(VI) removal as Fe species generate positive surface charges and redox-active sites that facilitate the reduction of Cr(VI) to Cr(III), while N groups enhance surface charge distribution and introduce addi-

tional binding modes, including coordination bonding, altogether intensifying electrostatic attraction [26].

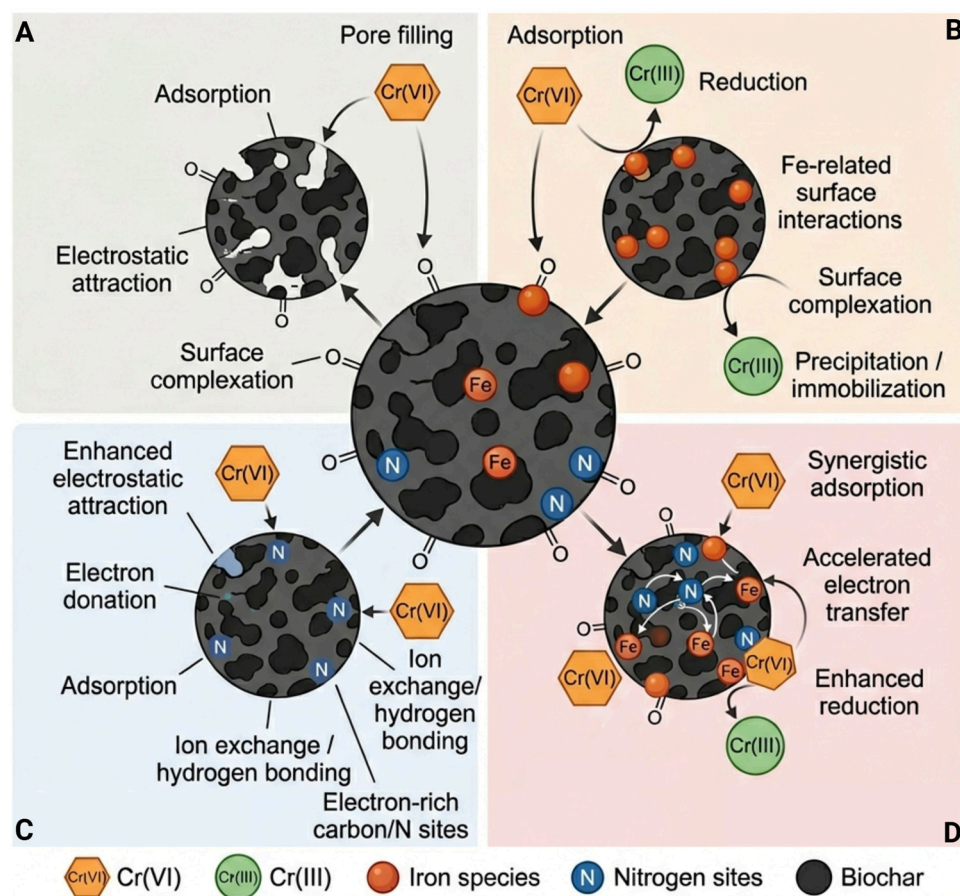


Figure 2. Mechanisms of Cr(VI) removal by pristine BC (A), non-noble metal-modified BC (Fe/BC: (B)), heteroatom-doped BC (N/BC: (C)), and non-noble metal and heteroatom co-doped BC (Fe-N co-doped BC: (D)).

8.1.2. Chemical Reduction

Chemical reduction plays the dominant and rate-limiting role by converting toxic Cr(VI) into less toxic Cr(III) under acidic conditions ($\approx 2\text{--}3$ pH) [46,106]. Cr removal mechanisms by Fe/BC involve combined electrostatic attraction, ion exchange, and chemical reduction of Cr(VI) to Cr(III) [117]. Fe-N/BC has an even more enhanced Cr removal performance due to N functionalities and highly reactive Fe NPs. Mechanistic analysis revealed dual contributions: partial physisorption on the BC surface and predominant chemical reduction of Cr(VI) to Cr(III) mediated by Fe(0) and Fe(II) species and chemical interactions strengthened by N doping, which enhances e^- transfer and surface active sites, thereby improving Cr(VI) reduction and adsorption efficiency [26]. The involvement of C=O and COOH groups in binding Cr species and the formation of Cr(III) oxide after adsorption indicate effective reduction. nZVI/BC exhibited a combination of physical sorption provided by hydrophilic BC and chemical reduction catalyzed by nZVI, resulting in rapid and highly efficient Cr(VI) removal [136].

8.1.3. Surface Complexation

Surface complexation during Cr removal occurs through abundant -OH groups, which can form surface complexes with Cr ions (Figure 2) [53,75]. A characterization confirmed that Fe_3O_4 acted as the key active site, facilitating chemisorption, ion exchange, and co-precipitation with Cr species. N/BC enhances Cr removal through surface complexation

by introducing amines and heterocyclic N into the BC matrix, boosting chemisorption and coordination interactions with Cr species [87]. Fe-N/BC attained a high adsorption capacity for Cr(VI) at acidic pH through surface complexation via N functional groups acting as e^- donors, facilitating the binding and reduction of Cr(VI). Surface complexation involves the formation of inner-sphere complexes between Cr species and non-noble metal and heteroatom co-doped BCs, further stabilized by heteroatoms providing additional binding sites [11].

8.1.4. Co-Precipitation

Fe/BC reduces Cr(VI) to Cr(III) by forming hydroxides such as $(Cr_xFe_{1-x})(OH)_3$ as precipitates, which co-precipitate onto the BC surface under neutral to alkaline pH, aided by BC functionalities [11]. Cr-laden BC is then separated by filtration or sedimentation, effectively immobilizing Cr. Fe/BC exhibited substantial improvements in Cr(VI) removal, primarily via adsorption, the reduction of Cr(VI) to Cr(III), and the co-precipitation of Cr-Fe species [76]. Similarly, nZVI/BC demonstrated co-precipitation, where Cr(VI) was converted to Cr(III) and precipitated with FeOx, such as $FeCr_2O_4$, improving removal while reducing passivation of Fe particles [137]. nZVI/BC also achieved a remarkably high Cr(VI) removal efficiency due to the effective dispersion of nZVI on BC surfaces and synergistic action promoting reduction and co-precipitation mechanisms [138]. PO_4^{3-} ions in F-P/BC also co-precipitated with Cr(III), further improving Cr removal efficiency [85].

8.1.5. Ion Exchange

Ion exchange works by increasing the number of BC active sites, which facilitates ion displacement (Figure 2) [27]. The removal of Cr(VI) using apple wood and glue residue BCs occurs partly through ion exchange as the main mechanism, alongside complexation, co-precipitation, electrostatic attraction, and reduction [52,53]. Similarly, Fe/BC derived from rice straw showed a substantial increase in Cr adsorption, corresponding to a 6.5-fold increase over that of pristine BC. In these BC materials, ion exchange was a contributing mechanism alongside chemical reduction and complexation [85]. Heteroatom doping combined with nano-zero-valent metal loading further enhances ion exchange capacity by introducing additional functional groups and increasing surface chemical activity. Ion exchange plays an important role in replacing surface-bound anions with Cr oxyanions, thus synergizing with reduction and co-precipitation to achieve enhanced removal [26,76]. The ion exchange efficacy of non-noble metal and heteroatom co-doped BCs can be improved by increasing the number of chemisorption sites, promoting e^- transfer, and making BC surfaces reactive and capable of exchanging Cr ions.

8.1.6. Pore Filling

Pore filling is significantly enhanced due to improved SSA, porosity, and active sites of BCs (Figure 2). $FeCl_3$ -modified BC showed improved physisorption through an increased SSA and pore volume, facilitating effective pore filling of Cr ions. BC achieved maximum adsorption capacity and maintained stability for repeated use, owing to its enhanced pore structure [39]. Non-noble metal and heteroatom co-doped BCs exemplify how metal modifications contribute to enhanced pore filling and physisorption due to the dispersion of nanoscale zero-valent metals within heteroatom-doped BC matrices, increasing Cr(VI) removal efficiency. Improved porous structures and surface chemistry increase the number of adsorption sites, allowing for effective physical uptake while facilitating other removal mechanisms [76]. Porous S-modified zero-valent Cu/BC showed better Cr(VI) adsorption, with pore filling as the dominant mechanism, accompanied by complexation, electrostatic attraction, and reduction [75]. Moreover, chicken manure-derived nZVI/BC demonstrated exceptional physisorption and chemical mechanisms. Well-dispersed Fe particles enhanced

BC porosity, facilitating Cr ion penetration and retention by pore filling and multilayer adsorption [139], reflecting the vital role of BC pore structure and Fe-based modifications in augmenting physisorption.

8.2. Characterization Techniques for BCs

Characterization techniques play a crucial role in verifying mechanisms of Cr removal by BCs before and after removal process [49,75]. Comprehensive analyses using scanning electron microscopy (SEM), X-ray diffraction (XRD), Raman spectroscopy, Fourier transform infrared spectroscopy (FTIR), and X-ray photoelectron spectroscopy (XPS) have been employed to elucidate Cr(VI) removal mechanisms [120,136]. SEM and XRD provide insights into morphology and crystalline structure changes caused by heteroatom doping and metal ion incorporation, while Raman spectroscopy helps identify graphitic carbon changes in BC matrix. To verify Cr(VI) to Cr(III) reduction on modified BCs, XPS (identifying Cr oxidation states and formation of Cr(III) oxide after Cr(VI) removal) and FTIR (confirming the involvement of C=O and COOH groups in e^- transfer and binding Cr species) are particularly instrumental in confirming presence of functional groups and redox transformation of Cr(VI) to Cr(III), which reveal a dual mechanism comprising adsorption onto BC surfaces and chemical reduction mediated by Fe(0) and Fe(II) species [26,136]. For instance, corn straw-derived Fe/BC was studied by SEM with energy-dispersive X-ray spectroscopy (SEM-EDS), XRD, FTIR, XPS, Brunauer–Emmett–Teller (BET) SSA analysis, zeta potential, and vibrating sample magnetometry (VSM) to unveil critical features: SEM-EDS showed surfaces where Cr was physically present, XRD identified iron oxide phases (e.g., $\gamma\text{-Fe}_2\text{O}_3$ and Fe_3O_4), FTIR detected surface functional groups involved in Cr binding, XPS provided Cr oxidation states and surface chemistry information, BET revealed SSA and porosity enhancements, and VSM assessed magnetic properties relevant to material recovery. The application of these techniques confirmed the Cr removal mechanisms of Fe/BC [140]. Additionally, FTIR, XPS, and SEM were used to verify the surface chemical composition and morphological changes in FeCl_3 -modified BC, where introduction of Fe-O groups was shown to coordinate with $\text{Cr}_2\text{O}_7^{2-}$ ions, enhancing adsorption sites and removal capacity [39].

8.3. Adsorption Isotherms and Kinetics

Isotherm and kinetic models are used to elucidate Cr removal mechanisms of BC. Langmuir, Freundlich, Sips, and Redlich–Peterson models are widely used for Cr(VI) removal isotherms [83,99,100], wherein Langmuir assumes homogeneous monolayer coverage, generally indicating chemisorption through specific binding energies, Freundlich describes heterogeneous surfaces and is often associated with physisorption on diverse sites, and Sips combines both Langmuir and Freundlich [75,109]. Among these, Langmuir is a better fitted isotherm for Cr(VI) removal [84,103]. Adsorption data for nZVI/N-BC closely fit the Langmuir model, indicating monolayer adsorption with a maximum capacity of 100.9 mg/g. Moreover, pseudo-1st-order (PFO), pseudo-2nd-order kinetic (PSO), and intraparticle diffusion (IPD) [66,79,80] models are widely used for Cr(VI) removal kinetics. In kinetics, PFO suggests physisorption dominated by diffusion, PSO indicates chemisorption involving e^- exchange/sharing, and IPD identifies rate-limiting steps [65,79]. Among these, PSO is a better fitted kinetic model for Cr(VI) removal [97,102], emphasizing that chemisorption predominates in Cr(VI) removal. Similarly, Fe/BC exhibited Cr adsorption, fitted the Langmuir and PSO models, and highlighted monolayer chemisorption as the rate-controlling step. BC showed an enhanced maximum capacity of 12.6 mg/g, reflecting the effect of Fe modification on adsorption affinity and surface interaction [85]. Additionally, nZVI/BC removal performance conformed to Freundlich and PSO models,

suggesting that heterogeneous surface adsorption and chemisorption dominated Cr uptake. The Avrami fractional-order kinetic model also showed a better fit for pristine and modified BCs, especially for Fe-N/PBC [27]. Collectively, Langmuir and PSO models consistently provide the best model fits with modified BCs, confirming that Cr(VI) removal mechanism is predominantly chemisorption with monolayer coverage governed by active surface sites introduced or enhanced by metal and heteroatom doping [112,113].

8.4. Computational Modelling

Density functional theory (DFT) and molecular dynamic (MD) simulations provide powerful computational frameworks for elucidating Cr removal mechanisms of BC at atomic and molecular levels. DFT and MD simulations eliminate the limitations of traditional trial-and-error experiments by providing atomic-resolution insights into exact mechanisms of binding, e^- transfer, and spatial orientation between Cr oxyanions and BC functional groups [141,142]. DFT is a quantum mechanical approach that enables a detailed investigation of electronic structures and interaction energies between Cr species and BC surfaces, capturing the nature of chemical bonding, charge transfer, and reduction processes crucial for Cr(VI) adsorption and transformation [143]. Applying DFT to carbon-based materials has demonstrated spontaneous and exothermic adsorption of HMs, offering a theoretical basis for interpreting the affinity of Cr ions toward functionalized BC surfaces [144]. When combined with MD simulations, these methods reveal dynamic structural adaptations, diffusion behaviors, and solvent effects influencing Cr adsorption and reduction kinetics. MD has been used effectively with DFT to evaluate surface interactions and stability in Fe-functionalized carbon systems, demonstrating how surface modifications affect adsorption energetics and molecular orientations [145]. Furthermore, DFT-based *ab initio* MD simulations allow for explicit capture of solvent and interface effects, which is essential for understanding the behavior of Cr species, with accurate predictions of adsorption geometries and reaction pathways [146]. These computational insights complement experimental characterizations such as XPS, XRD, and FTIR [147]. To elucidate molecular-level adsorption mechanisms of HMs on HCl-modified coconut shell BC, a nine-ring graphene surface model incorporating FTIR-derived $-OH$ and $-COOH$ groups was simulated with electrostatic potential distribution [148]. The $-OH$ and $-COOH$ moieties in electrostatic potential were identified as primary adsorption sites. In addition, force field parameter efforts grounded in DFT data enable enhanced MD simulations of relevant interfaces, such as surfactant or functional group interactions on FeO_x/BC , refining the prediction of adsorption equilibria and kinetics [145]. DFT calculations also verified that N and S co-doping decreased adsorption energy and increased attraction between modified BC and Cr(VI) through synergistic outcomes, consequently improving Cr(VI) adsorption [84]. Thus, synergy of DFT and MD simulations serves as a comprehensive theoretical toolkit for modelling and predicting Cr removal mechanisms through combined electronic structure calculations and atomic dynamics, ultimately guiding the optimization of modified BCs for efficient Cr removal.

8.5. ML for Optimizing BC Production for Cr Removal

The integration of ML techniques into studying Cr removal mechanisms using modified BCs can help address significant challenges and knowledge gaps in optimizing BC production, functional modification, and Cr removal [149]. Traditional experimental approaches for producing and modifying BC often involve complex parameters, including feedstock selection, pyrolysis temperature, doping agents, and impregnation conditions, where interactions can drastically influence physicochemical properties and Cr adsorption performance. ML offers powerful tools to navigate this multidimensional space

efficiently by modelling relationships between process variables and output properties or removal efficiencies, enabling predictive optimization from empirical trial-and-error methods to data-driven engineering [150]. For instance, artificial neural networks (ANNs) and adaptive neuro-fuzzy inference systems (ANFIS) have been used to predict Cr removal efficiencies from batch adsorption experiments, successfully capturing nonlinear effects of pH, adsorbent dose, and initial Cr concentration with high accuracy and robustness (Figure 3A,B) [151]. Such data-driven models facilitate the in silico exploration of parameter combinations, accelerating the identification of optimal production and operational conditions for maximum Cr uptake. SHAP analysis subsequently decodes ML frameworks, pinpointing how each variable (i.e., feedstock, pyrolysis parameters, and modification dose) influences the BC structure for targeted and real-world applications. For instance, SHAP values for the LSTM model (300 samples) relative to average removal capacities of Cr(VI) and Cr_{Total} are illustrated in Figure 3C and 3D, respectively. Primary predictors were Co, time, and T for Cr(VI) and Co, time, and pH for Cr_{Total} [149]. A higher Co, time, and T increased predictions for both, whereas an elevated pH decreased them. These variables impacted Cr(VI) more strongly than Cr_{Total}, reflecting the adsorbent mechanism: Cr_{Total} is constrained by limited biosorption sites compared to its higher capacity for Cr(VI) reduction.

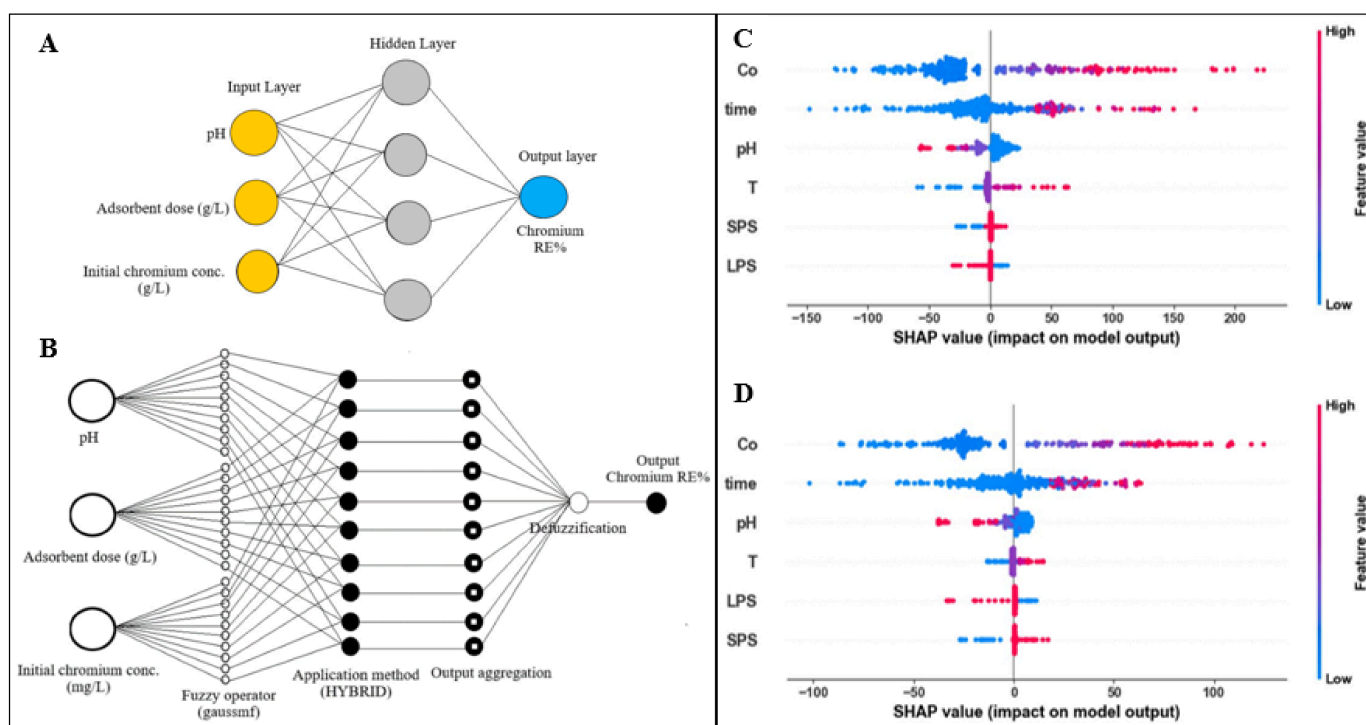


Figure 3. ANN (A) and ANFIS (B) frameworks for predicting Cr removal [151] and SHAP analysis: variable impact on removal capacities of Cr(VI) (C) and Cr_{Total} (D) (predictions for LSTM model for average Cr(VI) and Cr_{Total}) [149].

Moreover, physics-informed ML frameworks developed for metallurgical Cr systems illustrate how integrating domain knowledge with data-driven models can improve the interpretability and prediction of microstructural features influencing BC performance [152]. Adapting similar hybrid models for BC production could optimize precursor selection, doping levels, pyrolysis profiles, and post-treatments to tune porosity, surface functional groups, and redox-active sites for Cr removal. However, current limitations include the scarcity of large, high-quality, and standardized datasets explicitly linking BC production parameters to Cr adsorption capacity and durability, impeding reliable model training and

generalization. Ultimately, employing robust, interpretable ML frameworks to manage inherent variability of diverse biomasses and modification techniques will accelerate the development of cost-effective sustainable BC adsorbents, bridging experimental research with practical deployment in Cr-contaminated water treatment [151].

9. Performance Evaluation, Sustainability, and Economic Considerations

9.1. Batch vs. Column Studies

Batch tests are laboratory-based experiments conducted under ideal conditions (i.e., constant shaking, continuous liquid–solid contact, and complete mixing) that maximize Cr(VI) removal efficiency [97,113]. Utilizing inexpensive and abundant biomass feedstocks combined with waste-derived metal sources and industrial solid wastes (e.g., Fenton sludge and fly ash) to prepare N-doped mesoporous BC composites with large SSA and abundant O-containing functionalities yielded exceptionally high adsorption capacities (394 mg/g) and removal rates (80%) [153]. Column-based studies involving non-noble metals and BC showed maximum Cr(VI) adsorption capacities compared to pristine BC. A column study reported an adsorption capacity of 77.3 mg/g (93.6% efficiency) of Cr(VI) using magnetite NPs and HNO₃-modified BC [154]. However, in continuous columns, flow dynamics introduce severe limitations, e.g., axial dispersion reduces effective contact time and channeling and preferential flow bypass adsorption zones. Furthermore, real wastewater contains suspended solids and NOM, which physically clog pores and foul surfaces, drastically lowering dynamic capacity. Pressure drop increases with fine particles, limiting operational flow rates. Consequently, column capacities are often lower than batch equilibrium values.

9.2. Regeneration and Recyclability

Studies demonstrate that BC maintains significant Cr removal efficiency over multiple regeneration cycles, indicating its strong potential for reuse and cost-effectiveness (Table 6). Common regeneration solvents are NaOH, acids (HCl, HNO₃, and H₂SO₄), and their combinations. Modified BCs, such as CrFe₂O₄-loaded BC [155], ZVI/BC [156], N-rich AC [94], and N-P/BC [98], have been proven to be efficient for Cr(VI) removal, achieving higher removal efficiencies after multiple regeneration cycles. Furthermore, FeCl₃-modified BC exhibited Cr(VI) removal rate above 45% after five adsorption–desorption cycles, highlighting its durability and sustained adsorption performance in acidic conditions [39]. N-rich AC with abundant amine, pyrrolic, pyridinic, and protonated N groups showed excellent recyclability, with 99% Cr(VI) removal after 7th cycle [94]. The efficacy of regeneration varies with nature of adsorbent, adsorption mechanisms involved, and stability of active sites, which are often altered by modification processes [157]. Irreversible chemisorption, leaching of metal ions and surface functionalities, pore blockage, surface passivation, and structural collapse cause a reduction in BC efficiency for Cr(VI) removal. For example, Zn-doped NiFe₂O₄-pinecone BC achieved high Cr removal efficiencies but showed a gradual decrease in performance after three cycles, indicating the need for improved protocols to sustain long-term usability [22]. Nonetheless, regeneration of metal-modified BCs, especially those incorporating zero-valent metal NPs, faces challenges such as NPs oxidation, aggregation, and loss of redox-active sites during desorption and regeneration, which diminish adsorption capacity and increase costs [158]. This necessitates strategies to minimize secondary pollution from chemical regenerants and prevents BC structural degradation during regeneration to optimize lifecycle sustainability.

Table 6. Reusability of BCs for Cr removal.

BC	Run	Effluent Solution	Removal Performance	Ref.
N-P/BC	5	NaOH	Reduced from 96% to 75%	[98]
N/S co-doped BC	5	10% NaOH	Reduced to 77.3%	[159]
Fe/N-PBC	3	0.1 M NaOH	Reduced to 53%	[27]
nZVI/N co-doped BC	3	NR	Reduced to 57%	[26]
Fe/BC	5	0.1 M NaOH	Reduced to >45%	[39]
ZVI/BC	5	0.1 M NaOH	>80%	[156]
Fe/Mn oxide BC	6	4 M NaOH	Decreased to >30%	[158]
CS/Mn-Fe BC	5	NR	Decreased to 102.5 mg/g	[160]
CuO-MnO/BC	5	NR	Performance dropped to 8%	[135]
Zn/BC	6	0.5 M NaOH	Performance reduced from 84.2 to 59.7 mg/g	[157]
CrFe ₂ O ₄ -loaded BC	3	NR	90% efficiency	[155]
N-rich AC	7	NR	99% removal	[94]

Note: NR: not reported.

9.3. Energy Requirement and Carbon Footprint

The performance evaluation, sustainability, and economic considerations of Cr removal by BCs are closely linked to energy requirements and carbon footprint involved in their production. LCA reveals the environmental impacts of pristine and modified BC production, which are primarily dictated by feedstock selection, pyrolysis conditions, modification, and energy use against stable carbon storage, determining net sequestration and footprint [19]. Optimal BC production, characterized by high pyrolysis temperatures, efficient energy recovery, and minimized transportation, can result in net negative carbon emissions, with a median CO₂ removal of 1.4 kg CO₂-equivalent per kg of biomass [20]. In contrast, less efficient production methods that rely on fossil fuels and poor energy management may yield net positive emissions, detracting from sustainability goals. Metal modification introduces additional environmental impacts associated with metal extraction and wastewater generation, although these are generally minor compared to those from the pyrolysis process itself. Innovative pretreatments, such as hydrothermal carbonization, applied to lignin-rich biomass enhance BC thermal stability and increase hydrogen-rich syngas production while significantly lowering CO₂ emissions, thus improving overall energy efficiency and environmental production of BC [161]. The use of abundant and low-cost biomass waste and industrial residues combined with simple modification methods promotes environmental sustainability and cost efficiency by converting waste streams into value-added BC materials. The effective regeneration and reuse of BCs minimizes waste generation and raw material consumption, aligning with circular economy principles [162]. Utilizing renewable energy sources, integrating co-generated heat in BC production, and optimizing production to balance energy consumption with BC yield and functionality further reduce the carbon footprint and operational costs, supporting economic feasibility [163]. Integrated insights from LCA and techno-economic analyses underscore that energy demands and carbon footprints of producing modified BC can be minimized through technological improvements and green energy integration, thereby enabling environmentally sustainable and economically viable water treatment solutions [164].

9.4. Cost Comparison of Pristine and Modified BCs

Pristine BC generally incurs lower production costs, owing to simpler processing and fewer input materials; however, its efficacy in Cr removal tends to be limited by its lower adsorption capacity and lack of active sites. In contrast, metal modification notably enhances adsorption through increased active surface sites and redox capabilities [68], albeit with additional material and production costs associated with metal salts or NPs and modification procedures. The production of such BC composites generally involves complex and resource-intensive steps that elevate upfront costs but can be justified by higher Cr(VI) removal efficiencies and potential for regeneration. Similarly, BC composites such as CS-stabilized nZVI/BC achieved removal rates >97% at acidic pH, emphasizing their superior functionality despite probable higher production expenses [78]. Economic analyses suggest that although modified BC presents increased production costs, these can be offset by enhanced adsorption capacities, lower required doses, and prolonged material lifecycles stemming from improved regeneration potential [157,159]. Moreover, the use of biomass waste as feedstock and relatively low-cost metal sources favors environmental sustainability and cost-effectiveness compared to expensive conventional adsorbents [136].

10. Challenges, Knowledge Gaps, and Future Perspectives

Inconsistent performance across biomass types: Feedstock variability influences BC's SSA, pore structure, functional groups, and metal content. UV-modified BC derived from corn stalks exhibited superior Cr(VI) adsorption compared to those derived from sawdust or wheat straw, primarily due to increased O-containing surface functional groups and a larger SSA induced by UV treatment [21]. Similarly, Fe/BC derived from wheat straw, bamboo, and water hyacinth demonstrated variations in adsorption capacity (e.g., Fe/BC showed a higher capacity than pristine BC), which correlated with differences in Fe content, surface chemistry, and pore structure [117]. Moreover, heteroatom doping and metal loading synergistically enhanced Cr removal, and the extent of improvement was feedstock-dependent, highlighting the need for future studies focusing on agro-industrial waste to address waste management.

Challenges in scaling up: Metal and heteroatom co-doped BCs are widely used in controlled and laboratory environments for Cr removal, mechanisms, kinetics, and isotherms. However, replicating uniform doping and metal NPs dispersion consistently on an industrial-scale remains technically challenging, requiring precise control over reaction conditions and precursors. Specific challenges in scaling up include high-cost multi-step production of modified BCs, non-uniform metal–heteroatom co-doping at the ton scale, increased energy consumption, poor mechanical strength producing fines that clog fixed-bed columns, inadequate mass transfer and diffusion in continuous-flow and batch systems, rapid active-metal leaching deactivating sites, real-matrix fouling by organics/ions, and lack of sustainable and low-cost regeneration for spent BCs at field-scale [62,73]. These challenges hinder assessments of their long-term performance, stability, and regeneration potential during field-scale applications. Moreover, integration of these modified BCs into existing wastewater treatment infrastructures requires economic considerations, which involve balancing improved adsorption performance with higher production expenses, storage, and use of metal-modified BCs [156].

Environmental implications of modified BCs and NPs leaching: BC modified with functionalized NPs, especially BC-supported nZVI particles, is very effective for wastewater restoration [165,166], but metal oxidation/agglomeration and leaching of their particles also pose serious environmental risks. The major risks of leaching of metal NPs from metal-modified BCs during water treatment include secondary HMs contamination, nanotoxicity to aquatic ecosystems, ecological disruption, bioaccumulation and food chain transfer, and

potential transport of hazardous co-pollutants. Although BC NPs exhibit enhanced Cr(VI) adsorption and reduction capacities, concerns have arisen regarding the stability and mobility of embedded NPs under actual environmental conditions. Heat-treated hydrophilic BC impregnated with nZVI highlighted significant improvements in Cr removal, yet also underscores the possibility of NPs detachment and release into aquatic environments, posing ecotoxicological risks due to reactive Fe species or nano-sized materials [136]. Modified BCs derived from certain feedstocks or modified with metal salts may contain HMs, which may pose ecological risks. Moreover, co-pyrolysis and modification processes can inadvertently generate toxic by-products or alter BC structures, affecting microbial communities, fauna, or non-target organisms if released into the environment [167]. Additionally, environmental aging or interactions with biotic and abiotic factors can modify BC reactivity and contaminant binding stability, potentially leading to pollutant remobilization [167].

Competitive adsorption of co-existing pollutants: The adsorption behavior of Cr is affected by the presence of other pollutants. In real-world contaminated waters, numerous co-existing ions and pollutants, such as phosphate, ammonium, and HMs, can compete for adsorption sites, affecting removal efficiency and mechanistic pathways. Similarly, the presence of competing anions, such as SO_4^{2-} and PO_4^{3-} , was found to inhibit Cr(VI) adsorption onto FeCl_3 -modified BC, likely due to competition for positively charged sites or surface precipitation reactions [168]. Additionally, few studies have systematically evaluated competitive effects in mixed-metal or mixed-anion systems that better approximate environmental conditions.

Long-term stabilization of immobilized Cr(III): Stable immobilized Cr(III) after reduction of Cr(VI) remains a pivotal yet underexplored challenge in the study of Cr removal using modified BC. Although BCs effectively reduce Cr(VI) to Cr(III) via chemical reduction, adsorption, ion exchange, and co-precipitation, the persistence and immobilization of Cr(III) over extended periods under various environmental conditions require thorough investigation. FeCl_3 -modified BC demonstrated superior Cr(VI) adsorption and reduction efficiencies, attributable to Fe_3O_4 acting as key active species mediating Cr(VI) to Cr(III) transformation [168]. Nanoscale zero-valent metal-impregnated BCs exhibited enhanced Cr(VI) removal capabilities and formation of Cr(III) oxides associated with C=O and -COOH functional groups after adsorption [136]. The possible remobilization of Cr(III) under changing geochemical conditions (i.e., pH or *Eh*) could reverse immobilization and potentially release Cr back into the environment. Therefore, concerns remain regarding: (i) kinetics of Cr(III) reoxidation to Cr(VI) under natural fluctuating redox cycles; (ii) influence of competing organics (e.g., humic acids) on Cr(III) complexation and mobilization; (iii) aging behavior of BCs that destabilizes retained Cr; (iv) role of microbial activity in mediating Cr speciation; and (v) lack of standardized long-term field leaching and accelerated aging protocols to predict decade-long performance, hindering regulatory acceptance and safe disposal pathways.

11. Conclusions

This review critically evaluates the performance of pristine and modified BCs (non-noble metal/BC, heteroatom/BC, and non-noble metal-heteroatom/BC) for Cr(VI) removal from contaminated water. Pristine BC exhibits limited adsorption capacity, poor surface functionality, and inadequate redox activities. In contrast, modified BCs significantly enhance Cr(VI) removal through synergistic and enhanced mechanisms, including electrostatic attraction, reduction to less toxic Cr(III), surface complexation, ion exchange, co-precipitation, and pore filling. Metal modification of BC introduces magnetic properties for easy separation and abundant reactive sites, while heteroatom doping enriches surface functional groups and improves e^- transfer. The combination of non-noble metal and

heteroatom yields superior adsorption performance, far exceeding that of pristine BC. Removal efficiency is strongly pH-dependent, with optimal performance under acidic conditions ($\approx 2\text{--}3$ pH), as confirmed through characterization techniques such as SEM, FTIR, and XPS and modelling approaches (Langmuir isotherms, PSO kinetics, and DFT calculations). Despite promising laboratory-scale results, several challenges remain, including inconsistent performance across biomass types, limited field-scale validation, potential environmental risks from NPs leaching, competitive adsorption in co-existing pollutant systems, and uncertainty regarding the long-term stability of immobilized Cr(III). Therefore, future research should focus on cost-effective regeneration methods, LCA, and application of ML to optimize BC production and modification, which will facilitate the practical deployment of modified BCs as sustainable, efficient, and economically viable solutions for Cr(VI) removal in real-world water treatment applications.

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